



ADMINISTRATOR MANUAL · ENGLISH EDITION

Business Process Map Administrator Manual

Consoles, operations and governance for system administrators. Everything in the User Manual applies to you too.

Company process-map system • As of 2026-09-30

Navigate with   or click the screen edges ·  to restart
Business Process Map · Administrator Manual · 2026-10-02

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01

The Sysadmin Role

How sysadmin is granted and what it unlocks.

How it's granted

Environment variable

Not in the DB. `BPM_SYSADMINS` (comma-separated login IDs), read at startup.

Dev default

With auth off, effectively everyone is sysadmin.

Real auth

With `DEV_ENFORCE_PERMISSIONS=true` or real auth, listed IDs only.

Idap mode

The Local Accounts Sysadmin toggle can grant it too (effective only there).

Employees tag

Settings → Directory → Employees tags them Sysadmin. `Env-listed users` can't be turned off from any screen.

What sysadmin unlocks

Owner on every map

Collaborators · approvers · visibility · versions · deletion.

Admin consoles

Every admin-only console under Settings.

Withdraw is the exception

Pending / Approved versions only by the submitter. Rejected also by owner / sysadmin.

Force checkout

The only role that can take another user's edit lock.

Moderation

Group approvals · feedback replies · the global trash · manual publishing.

02

Admin Console Map

Every console on the Settings rail, at a glance.

The Settings rail

Content

Notices · Manual · **Knowledge base** · AI chat · **AI prompts** .

Directory

Employees · Departments · **Catalogs** · **Orphaned refs** · Local Accounts (Idap only).

Framework

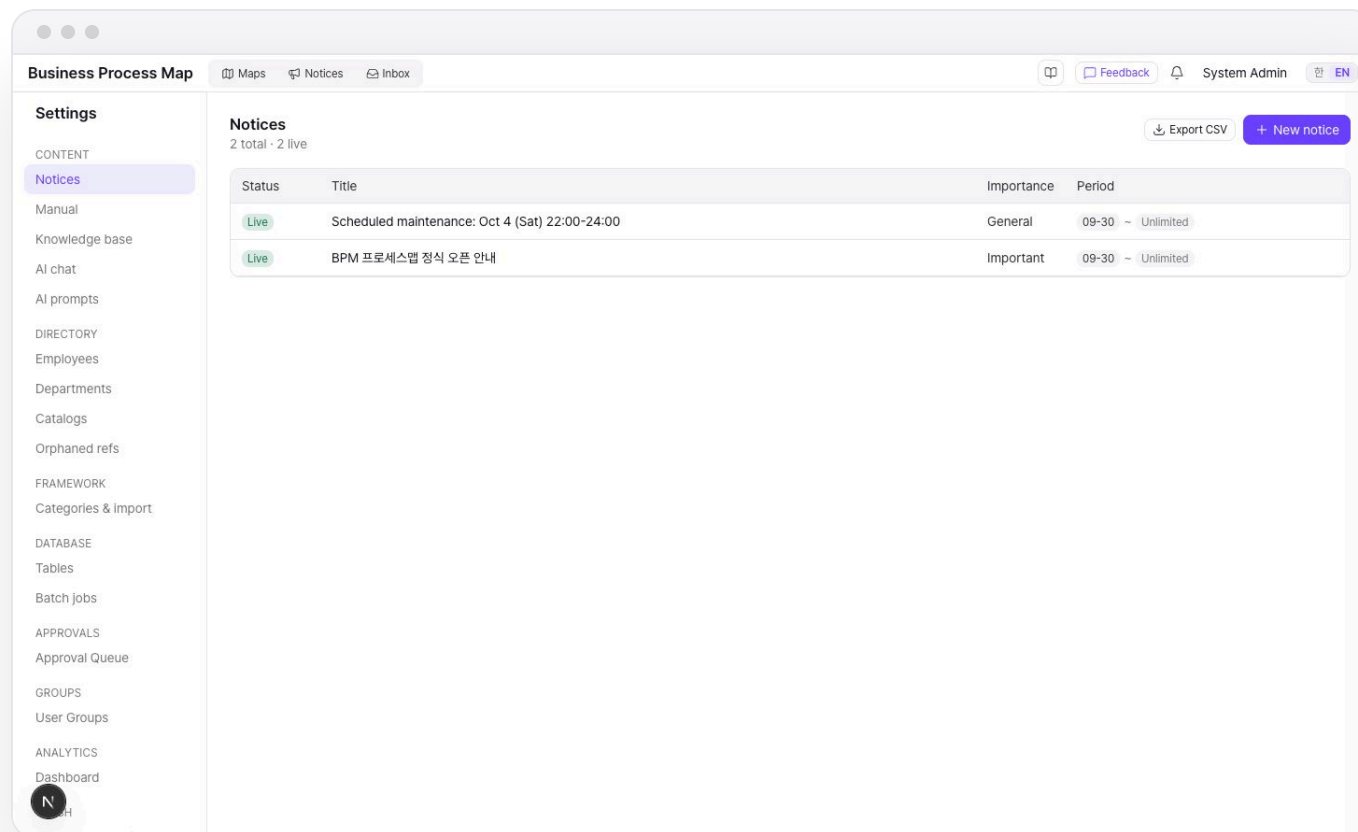
Categories & import (**Manage / Status** views). Delegated admins see their subtree.

Database · Approvals

Tables · Batch jobs (**Back up now**) · Approval Queue.

More

Groups · Analytics (Dashboard) · Trash (Scheduled deletion).



03

Content Management

Notices, the in-app manual, AI knowledge · settings · prompts, and feedback.

Notices

Fields

Title · importance (Important / General) · posting period (**open-ended** allowed) · Markdown body.

Notify all

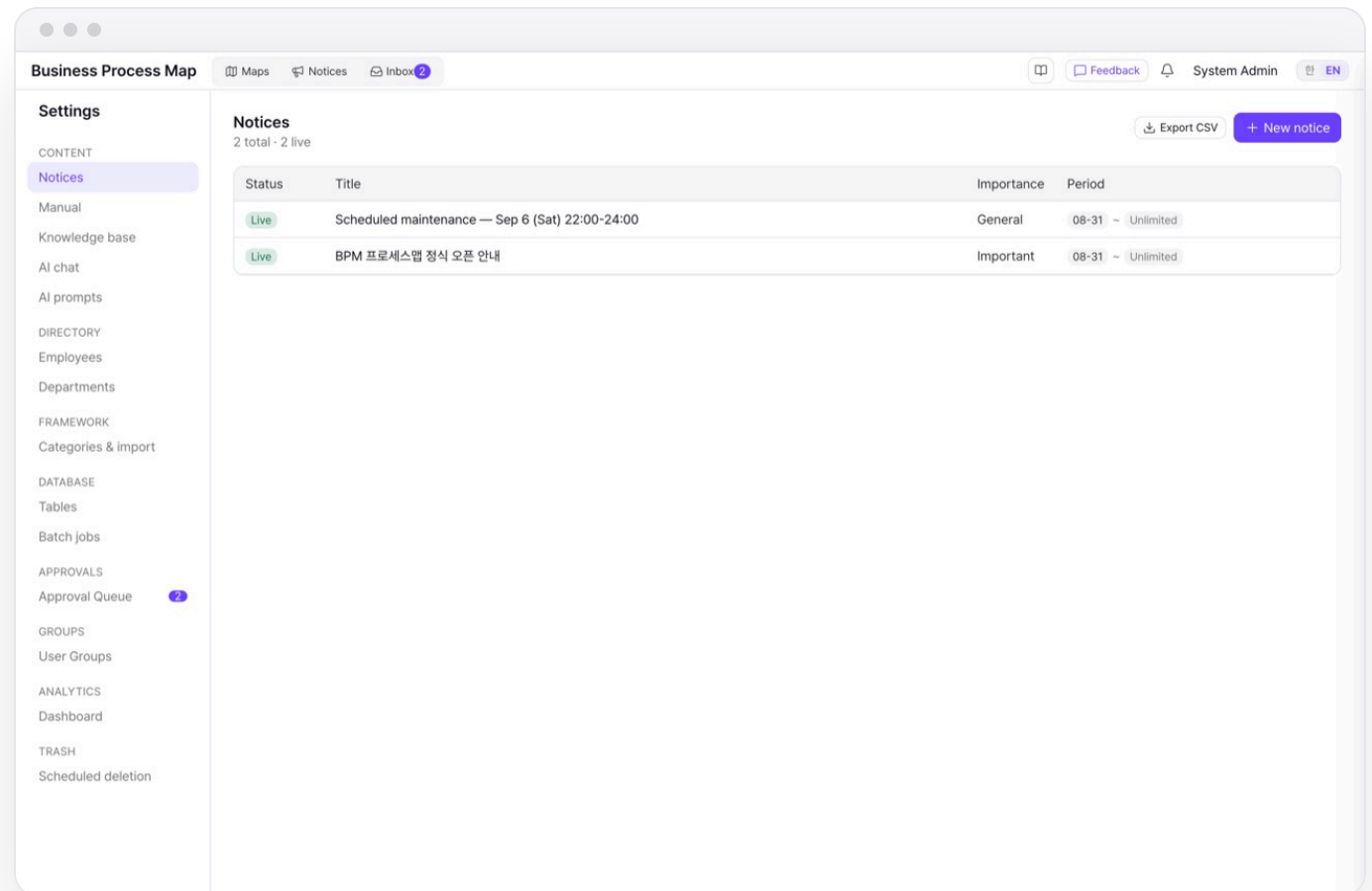
Check **Notify all users on publish** to push to every inbox.

Visibility

Users see active notices only, the admin list shows all.
CSV export.

Hard delete

Permanent, no trash. Already-sent bell notifications remain.



Publishing the manual

Several documents

`/manual` . Each with a format (markdown / sanitized HTML) and language (KO / EN).

Authoring

Write inline · upload an .md · load the bundled copy.
Preview, then publish.

KO / EN pairing

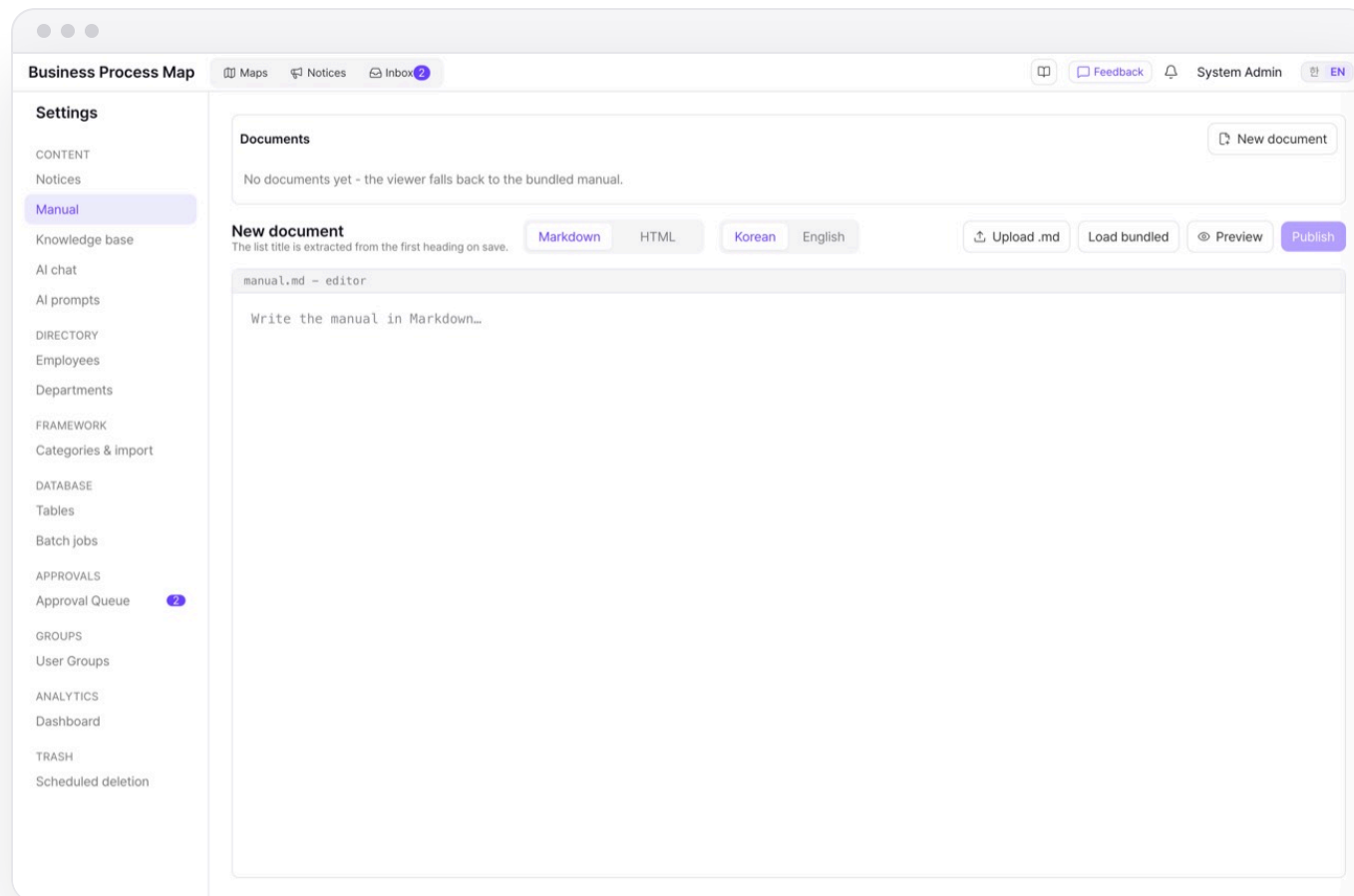
Titles come from the first heading. Same order keeps the language toggle on the same document.

Fallback

With no publish history the bundled copy is served with a badge.

TOC rule

`##` / `###` headings. Images and raw HTML don't render.



Knowledge base

Purpose

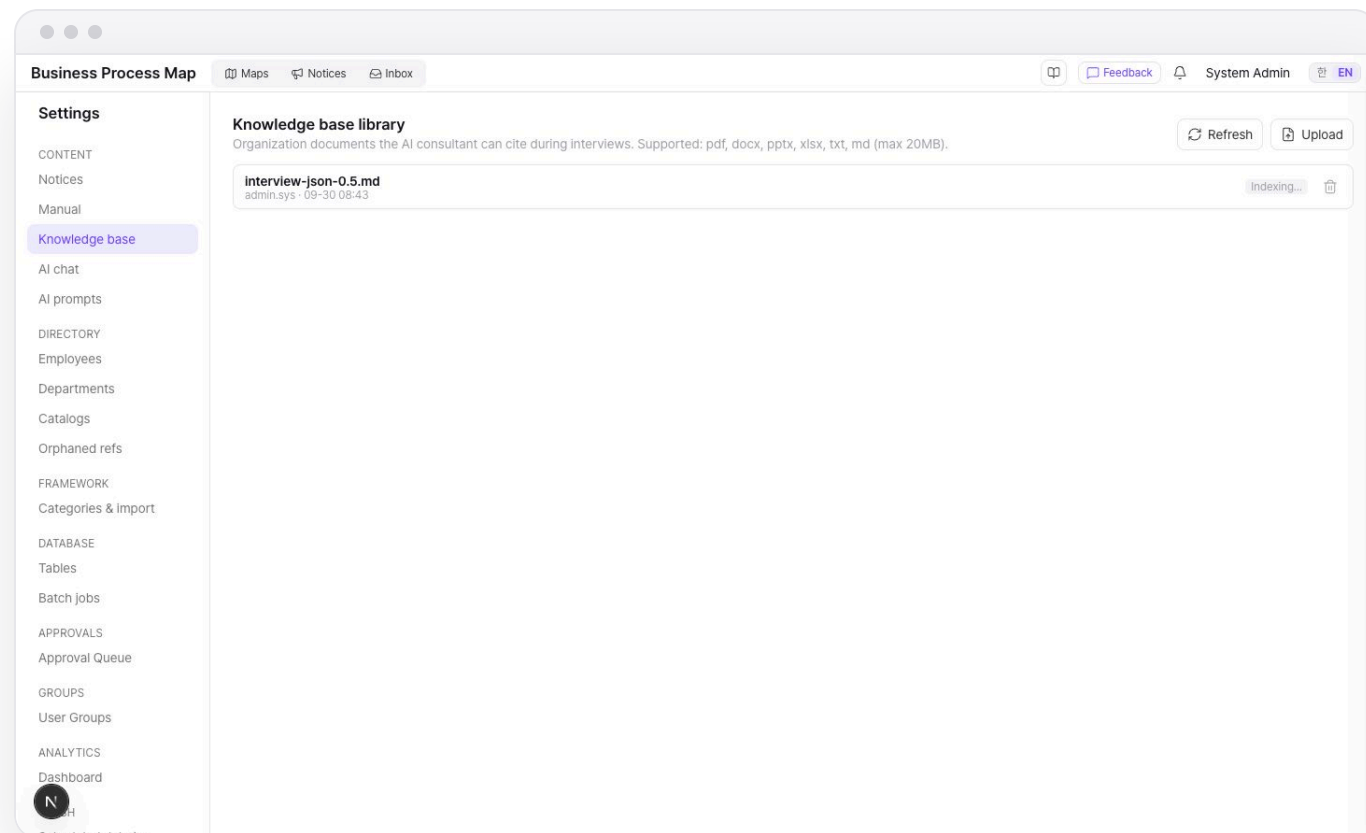
Org documents the AI consultant cites in interviews.
Upload SOPs and guides.

Formats

pdf · docx · pptx · xlsx · txt · md, up to 20 MB per file.

Prerequisite

Works only with both `AI_ENABLED` and the embedding server `EMBED_URL` set.



AI chat settings · AI prompts

AI access switch

Pauses every AI feature without a redeploy. Availability =

AI_ENABLED AND the switch.

Chat storage

Always server-side (user × map, owner-visible). Not a toggle.

Retention caps

20 chats per map · 200 messages per chat · 180 days.

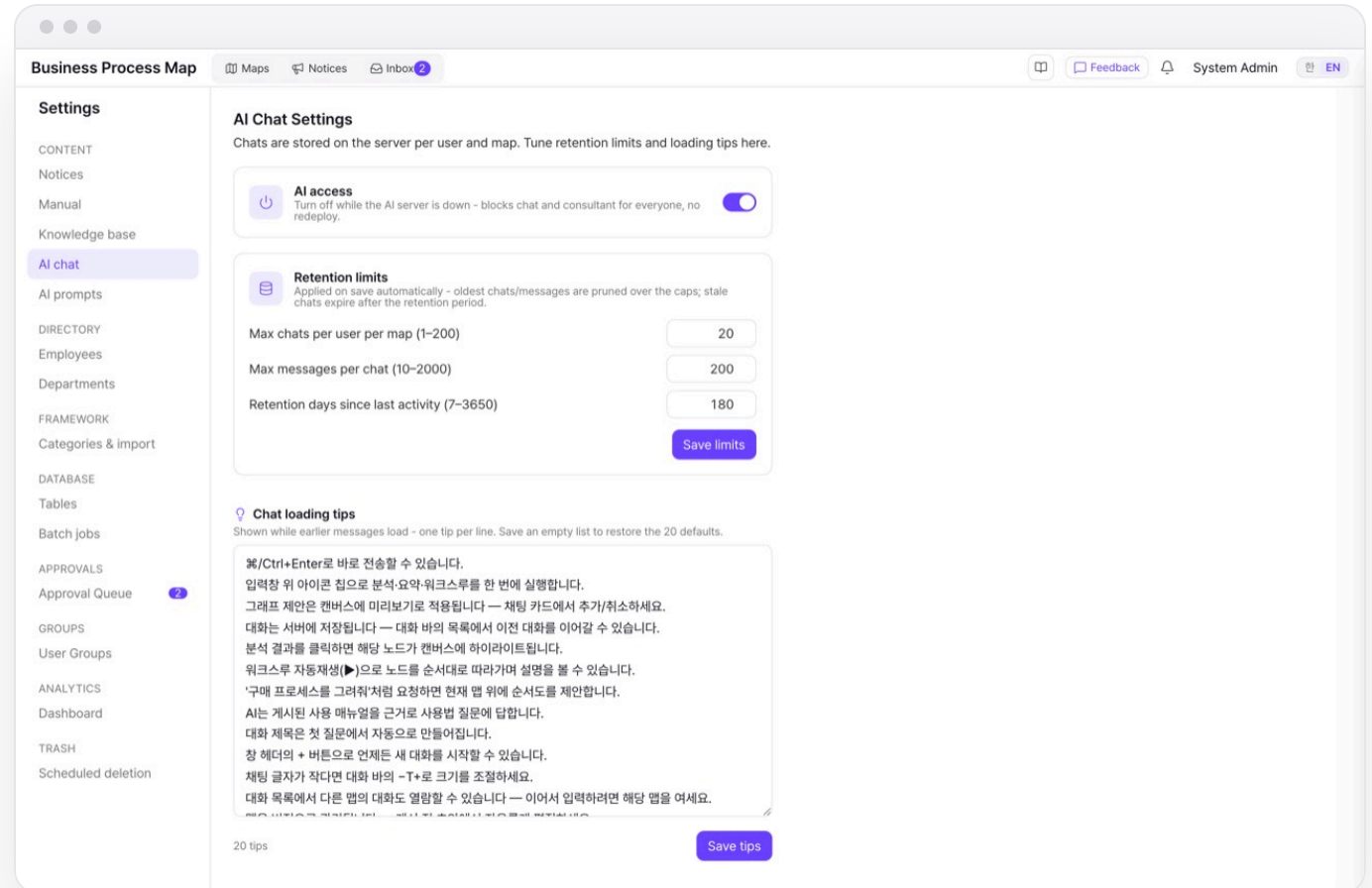
Applies immediately.

AI prompts

Override the built-ins. Saves apply instantly, clearing falls back.

Re-check

While overridden, prompt improvements shipped in code are not picked up.



Feedback administration

Status · replies

Sysadmin only. **Done** locks further replies.

Deliberate notifications

Send notification (reply, resendable) ·

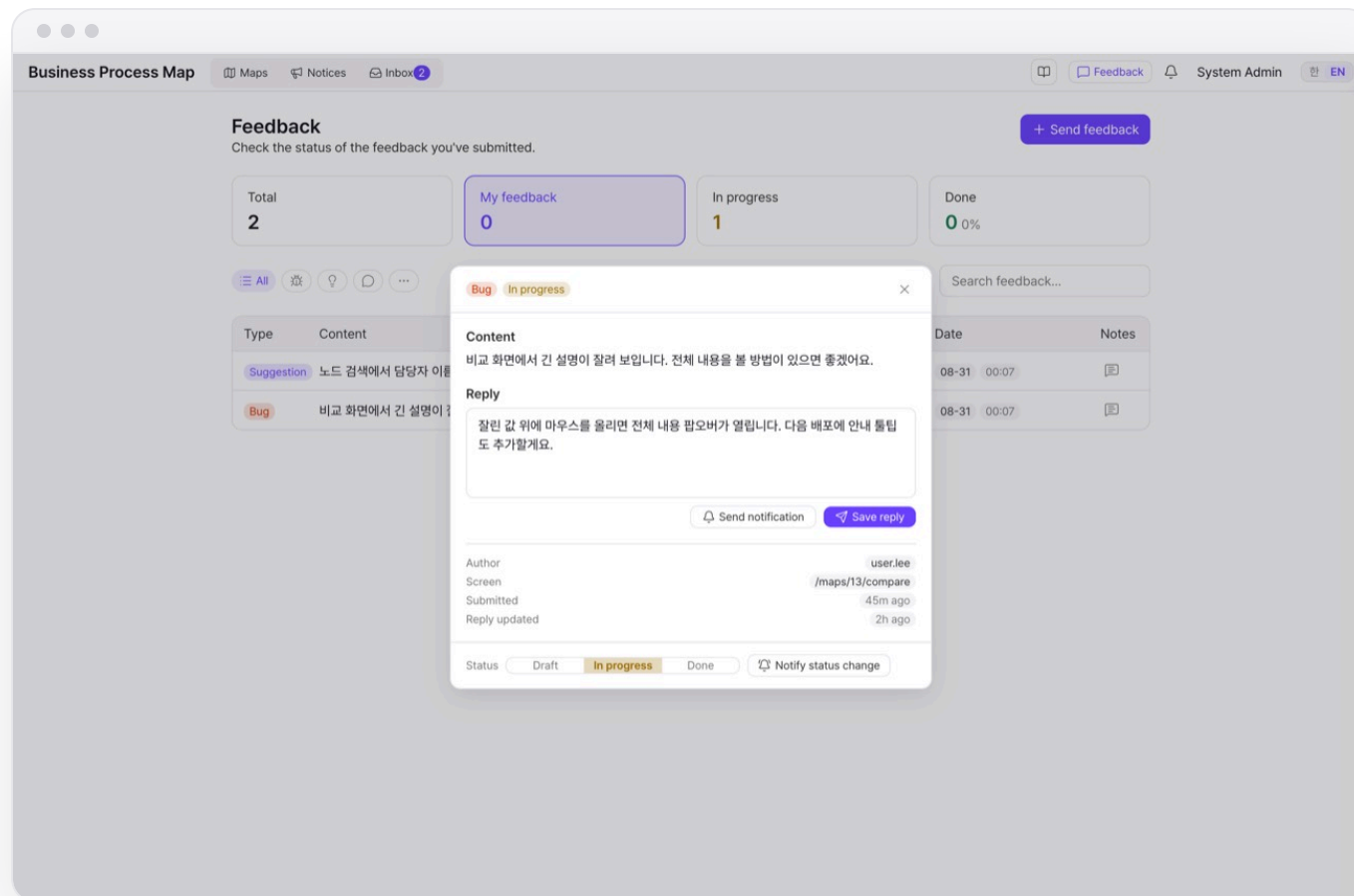
Notify status change (once).

Sent log

The "Notification sent {time}" meta row.

Notes

Anyone writes, edits keep history, delete = archive. Purge only via the `feedback_notes` table.



04

Directory & Org

HR-webhook sync, the departments basis, heads, and consultant accounts.

Employees & sync

Source

The n8n HR webhook. A full sync refreshes employees plus departments (AD only enriches titles).

Sync all from AD button

Employees tab. Ends with a scanned / upserted / deactivated / deleted summary. One per 5 min.

Auto sync

`HR_SYNC_INTERVAL_HOURS` (default 24 h) plus per-user daily sync on login.

Safeguards

The sync-preview dry run and the deletion cap

`HR_SYNC_DELETE_CAP_PCT` (default 20%).

Departures

Not deleted. Marked active=false and excluded from pickers.

Business Process Map | Maps | Notices | Inbox 2 | Feedback | System Admin | 한 EN

Settings

- CONTENT
 - Notices
 - Manual
 - Knowledge base
 - AI chat
 - AI prompts
- DIRECTORY**
 - Employees**
 - Departments
- FRAMEWORK
 - Categories & import
- DATABASE
 - Tables
 - Batch jobs
- APPROVALS
 - Approval Queue 2
- GROUPS
 - User Groups
- ANALYTICS
 - Dashboard
- TRASH
 - Scheduled deletion

Employees | Export CSV | Sync all from AD

Exposed positions
Choose which EDW titles are shown as a person's position across the app.

☒ 그룹장 ☒ 파트장 ☒ 팀장 ☒ 센터장

Save

loginId	name	korean name	korean dept	title	department	role	status	source
admin.sys	System Admin	Sysadmin		Manager	System Team	admin	Active	local
bora.choi	Bora Choi			Lead	Support Team	user	Active	local
bora.hong	Bora Hong			Lead	Brand Part 1	user	Active	local
bora.hwang	Bora Hwang			Manager	Planning Part 1	user	Active	local
bora.jang	Bora Jang			Manager	Brand Part 2	user	Active	local
bora.jung	Bora Jung			Manager	Growth Team	user	Active	local
bora.jung2	Bora Jung			Senior	Brand Part 3	user	Active	local
bora.kwon	Bora Kwon			Lead	Brand Part 2	user	Active	local
bora.lee	Bora Lee			Senior	Planning Part 1	user	Active	local
bora.lee2	Bora Lee			Senior	Facilities Office	user	Active	local
bora.lee3	Bora Lee			Associate	Planning Part 1	user	Active	local
bora.oh	Bora Oh			Associate	Insights Team	user	Active	local
bora.oh2	Bora Oh			Staff	Planning Part 2	user	Active	local
bora.park	Bora Park			Senior	Logistics Part 1	user	Active	local
bora.seo	Bora Seo			Senior	Analytics Part 1	user	Active	local

Departments & heads

Org basis

The departments table is the single source for the org chart and path resolution.

Heads

Determined from EDW position (FRNM) data, tagged Manager on the chain.

Exposed positions

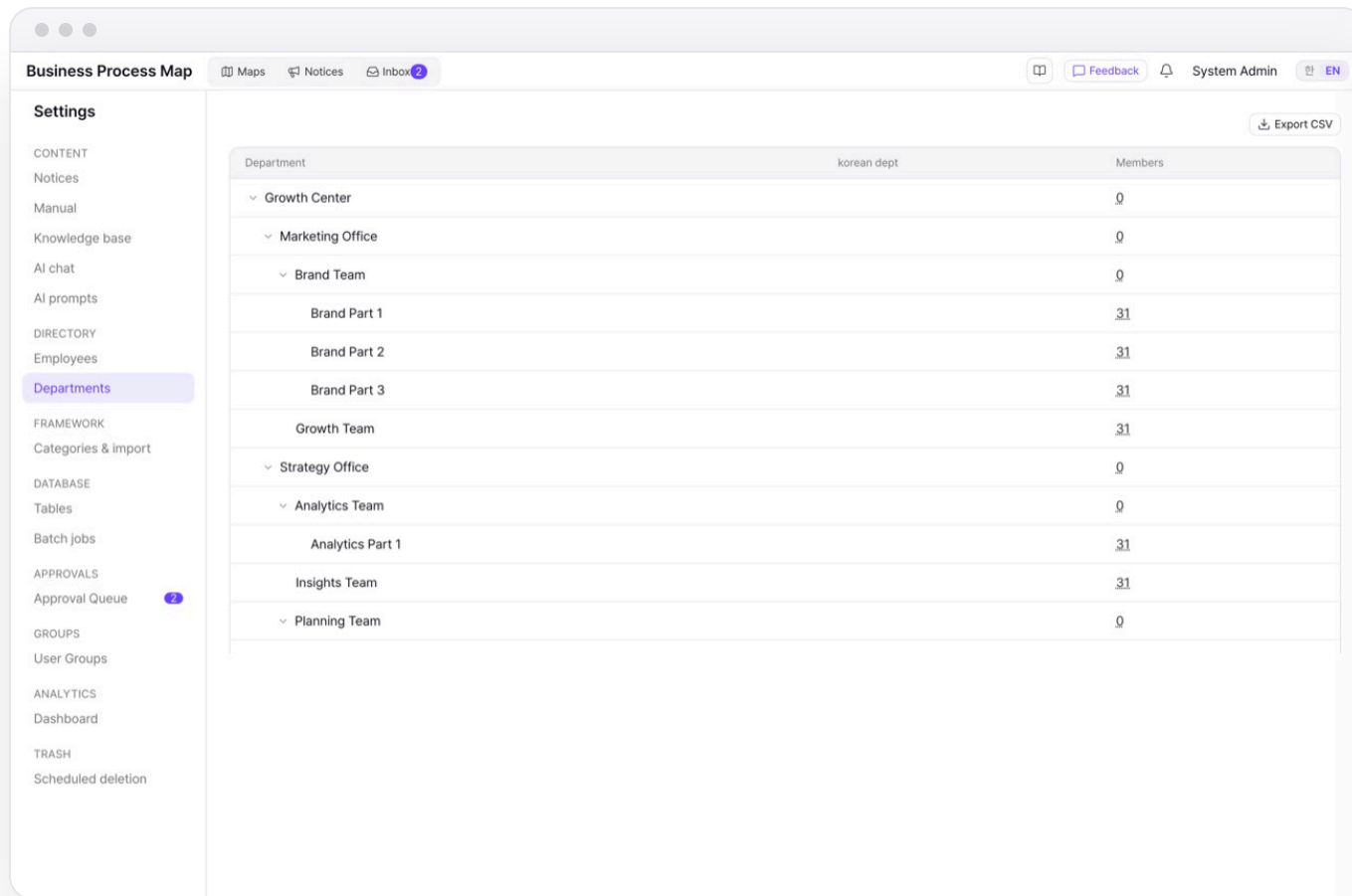
An Employees-tab card picks which EDW titles show (defaults: 그룹장 · 파트장 · 팀장 · 센터장).

Clean-up

Stale department and departed-person references go to the **Orphaned refs** tab.

CSV

Both tables export (BOM, formula-injection guard).



The screenshot shows the 'Business Process Map' application interface. On the left is a sidebar menu with categories like CONTENT, FRAMEWORK, DATABASE, APPROVALS, GROUPS, ANALYTICS, and TRASH. The 'Departments' item under the DIRECTORY category is selected. The main area displays a table with columns 'Department', 'korean dept', and 'Members'. The table lists various departments and their member counts. An 'Export CSV' button is in the top right corner of the table area.

Department	korean dept	Members
▼ Growth Center		0
▼ Marketing Office		0
▼ Brand Team		0
Brand Part 1		31
Brand Part 2		31
Brand Part 3		31
Growth Team		31
▼ Strategy Office		0
▼ Analytics Team		0
Analytics Part 1		31
Insights Team		31
▼ Planning Team		0

Catalogs (Roles & Systems)

Autocomplete lists

For node Role and System. **Role | System** tabs, item list left, **alias** editor right.

Aliases

Typing one saves the canonical spelling (`sap` → SAP ERP). One alias belongs to one item.

CSV import

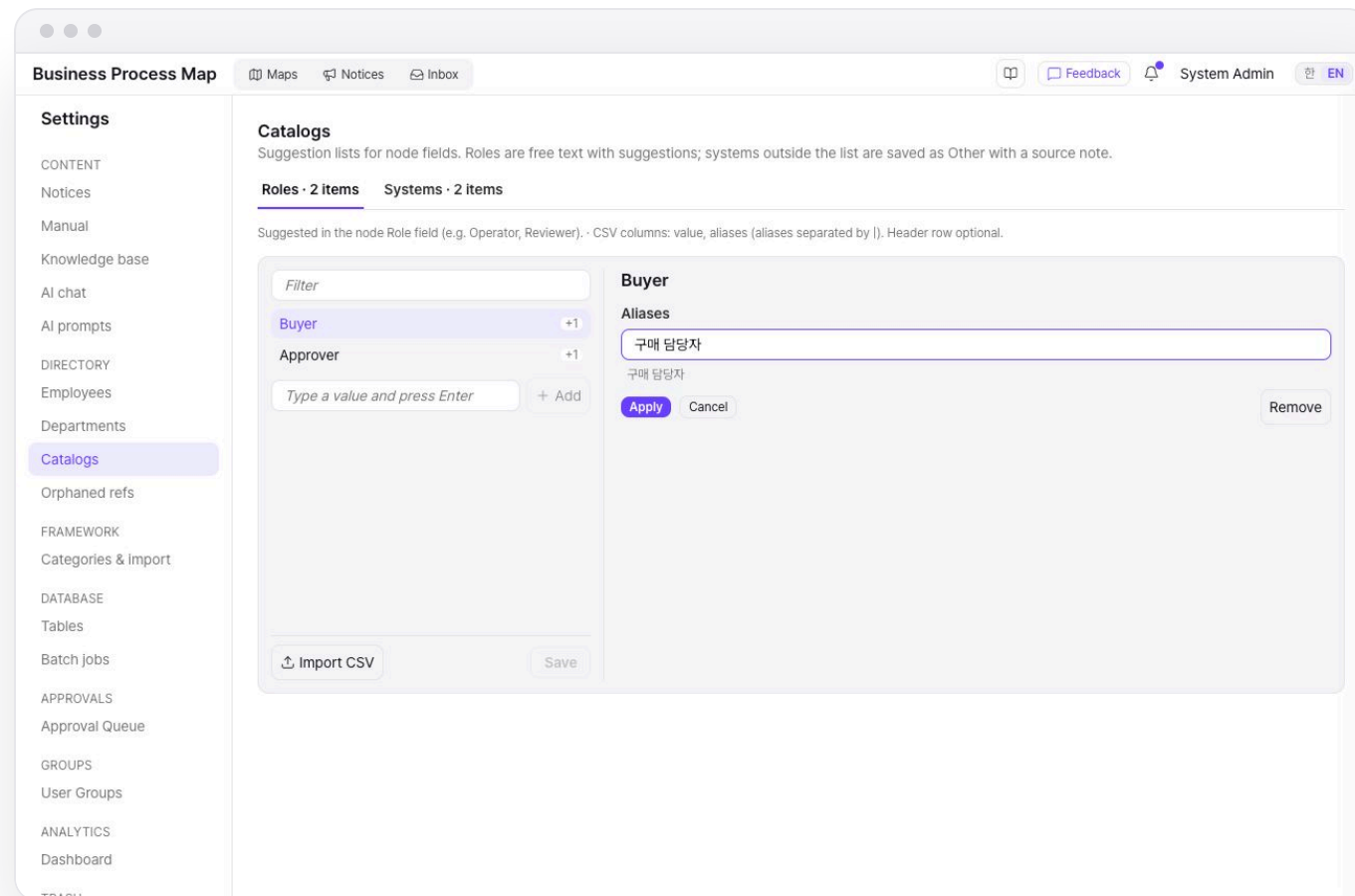
Two-column `value,aliases` (split by `|`). The System tab promotes values in use.

Other

Reserved in the System list (undeletable, always on top).
Editing is sysadmin-only.

Scope

Editor · CSV · AI assistant · interview · JSON import all normalize against these lists.



Orphaned reference audit

Rescan

12 places on demand. Node departments / assignees, SP designations, owners, collaborators, approvers, groups, admins.

Grouped by value

Vanished department path or departed person. Expand for each reference with a checkbox.

Bulk replace · remove

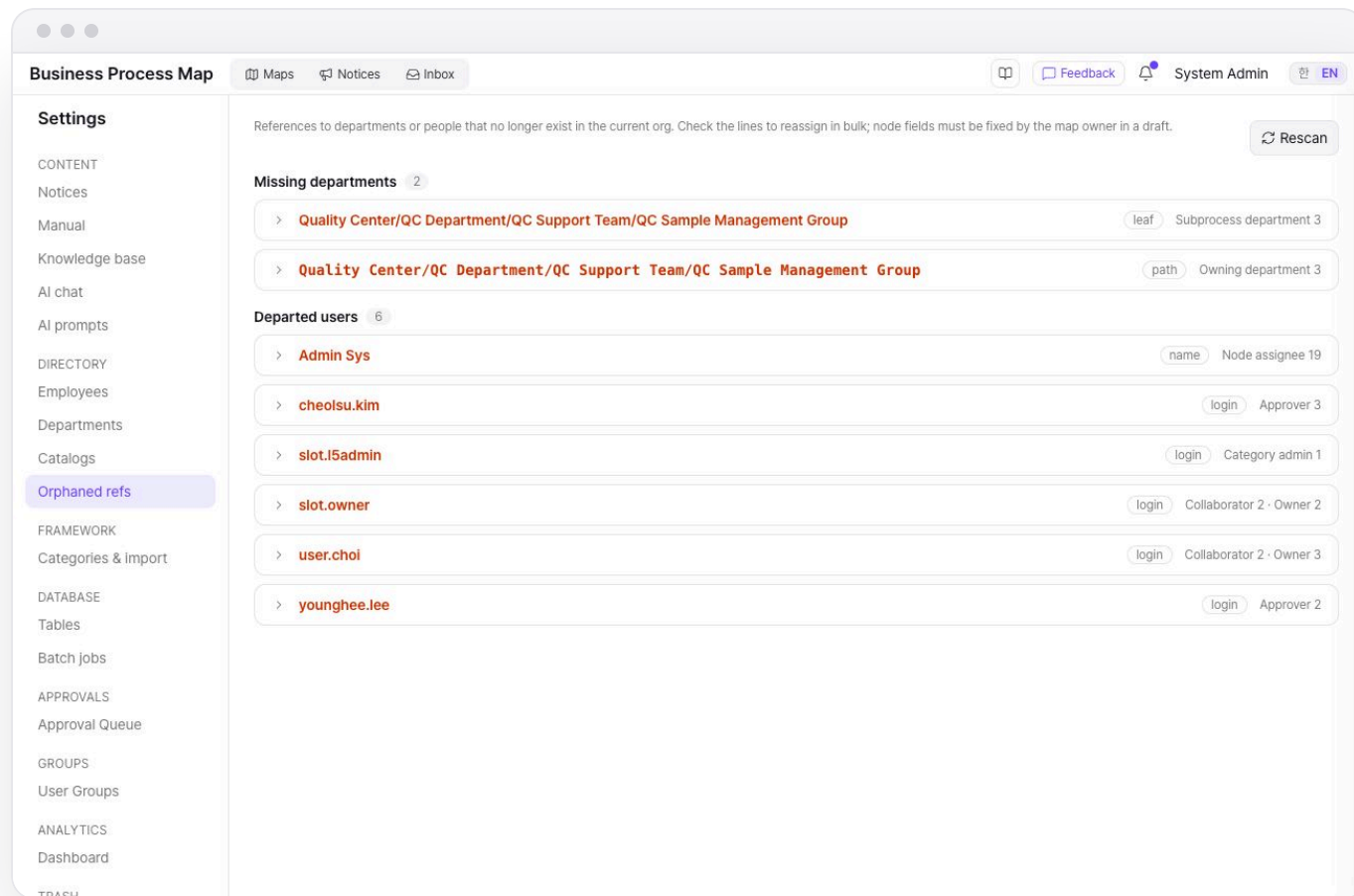
Checked lines get a current department or person. Node fields are display-only.

Owner notices

Stale refs fix request (one per owner). Departed owners get a new one assigned.

User side

Stale refs badge on cards, the home Issues filter, warning-colored node icons.



Local accounts (Idap mode)

When shown

Only in Idap auth mode. ID / password logins for external consultants.

Per account

Reset password · deactivate / reactivate · delete (permanent). Deactivate first when an engagement ends.

Tokens

Lifetime and secret rotation live in docs/deploy/deploy.md §2.1.

Sign-in order

Local accounts first, then AD fallback. 5 attempts per 5 minutes.

Sysadmin toggle

Local accounts only . Directory accounts only via `BPM_SYSADMINS` .

05

Framework Management

Category tree and level delegation, linkage admins, confirmation governance, slot changes, the status board, interview import, filling an L5 with AI.

Category management & level delegation

Layout

Manage / Status toggle. Tree left, detail panel right.
Single-child chains open through to the fork.

Six actions

Add child · Rename · Managing dept · Linkage admins ·
Move · Delete. Blocked ones show a reason.

Rules

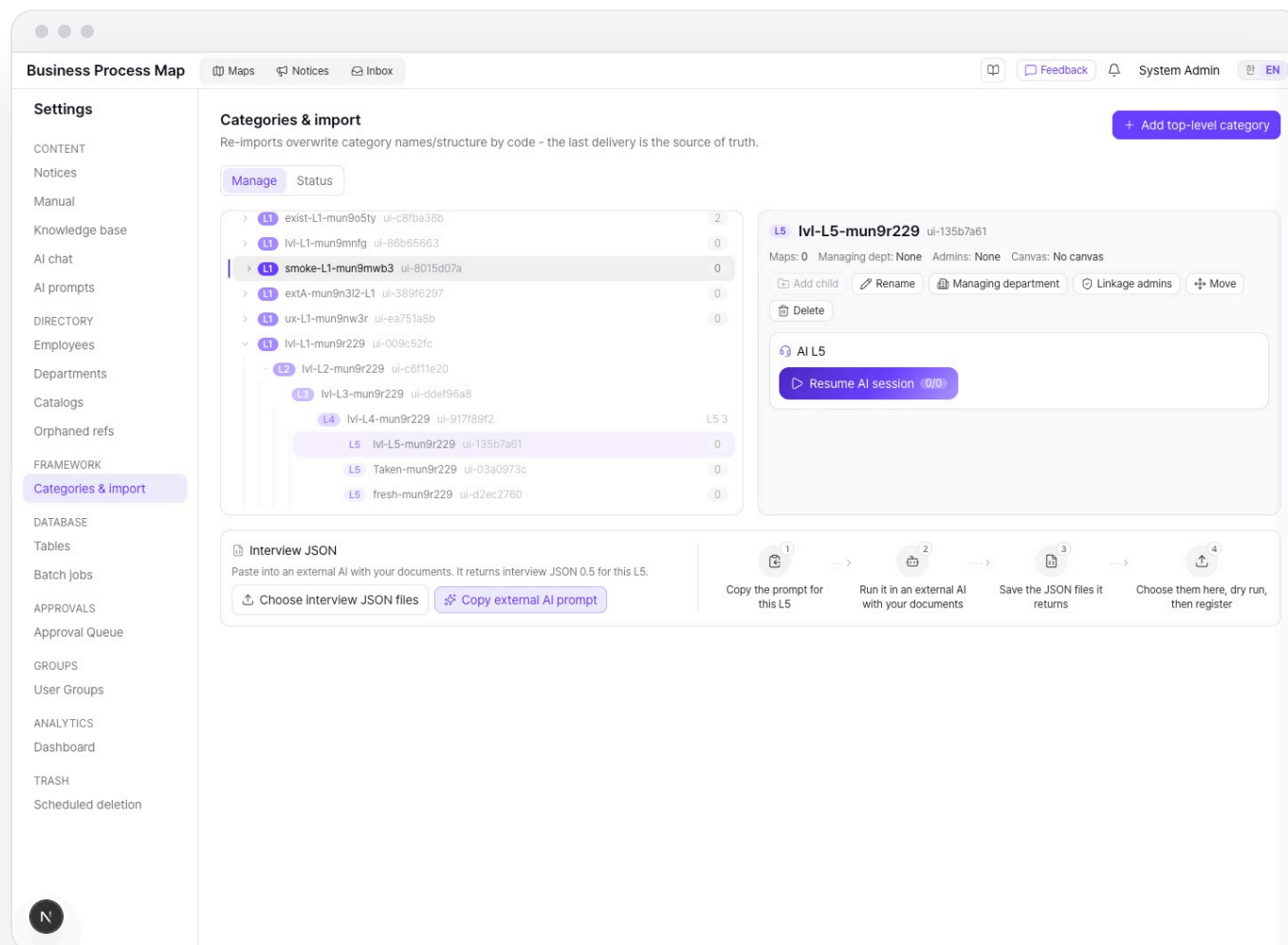
Max 5 levels, maps on L5 only, no duplicate siblings (409).

Managing department

Empty inherits from the parent. Becomes the canvas's
owning department.

Delegation

Admins add / rename / reorder under their own category.
Can't move or delete it.



Linkage admins

Appoint

Select a row → **Linkage admins** in the detail panel. Users / groups, buffered until one confirm.

Inherit downward

Appoint at L1–L4 and every L5 canvas below becomes editable.

Shown as

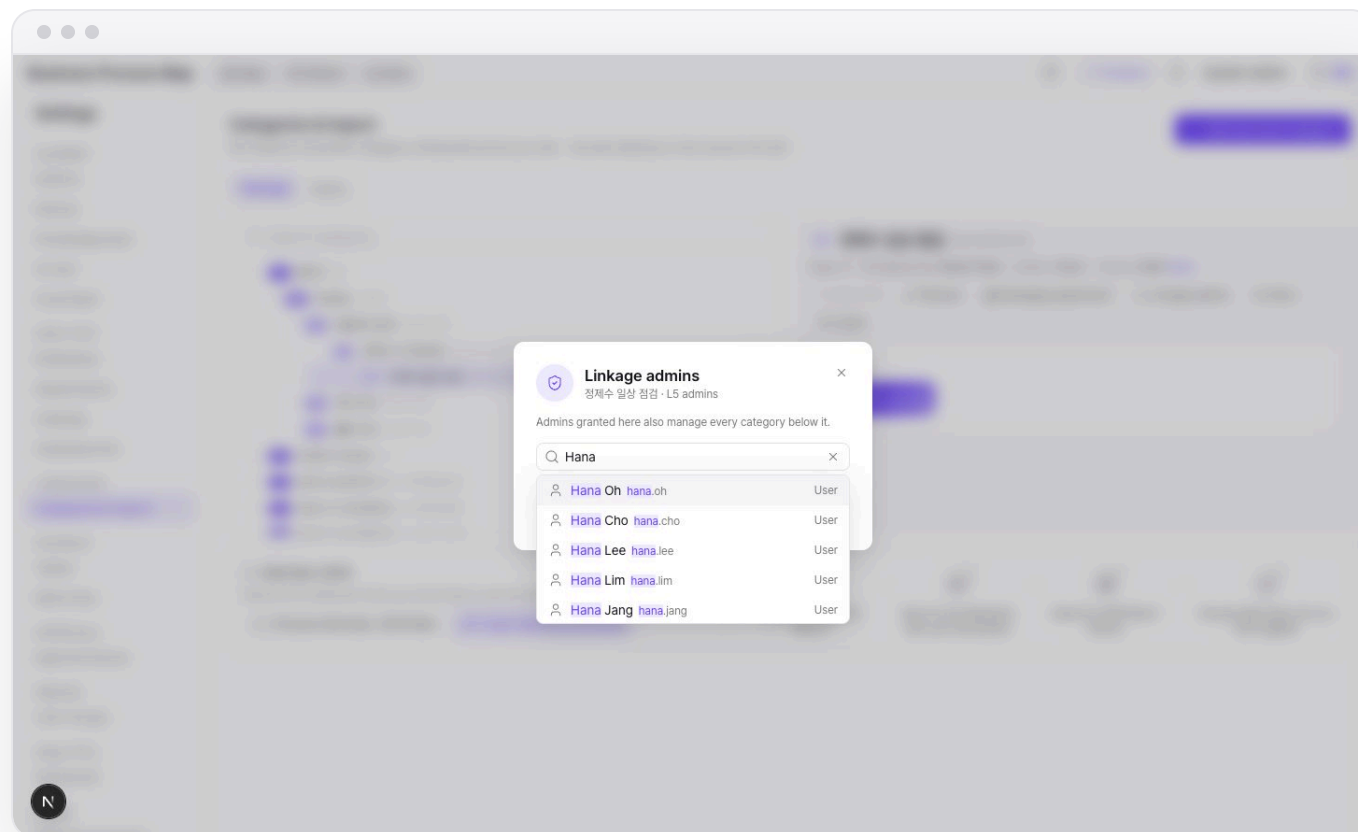
The detail panel's info line (three names, then +N), dual-named per language.

Canvases

Real maps (`mode=framework`) hidden from the map list and SP library.

Major confirm

Permanently prunes the previous major's intermediate minors (X.0 and the last minor stay).



Confirmation governance

confirmed state

Apart from publishing. Snapshots and the draft can't be deleted. Regular workflows are blocked server-side.

Six gates

All L6 placed · no placeholders · no stale links · all L6 published · no exit-less loops · branches use a decision.

Single source

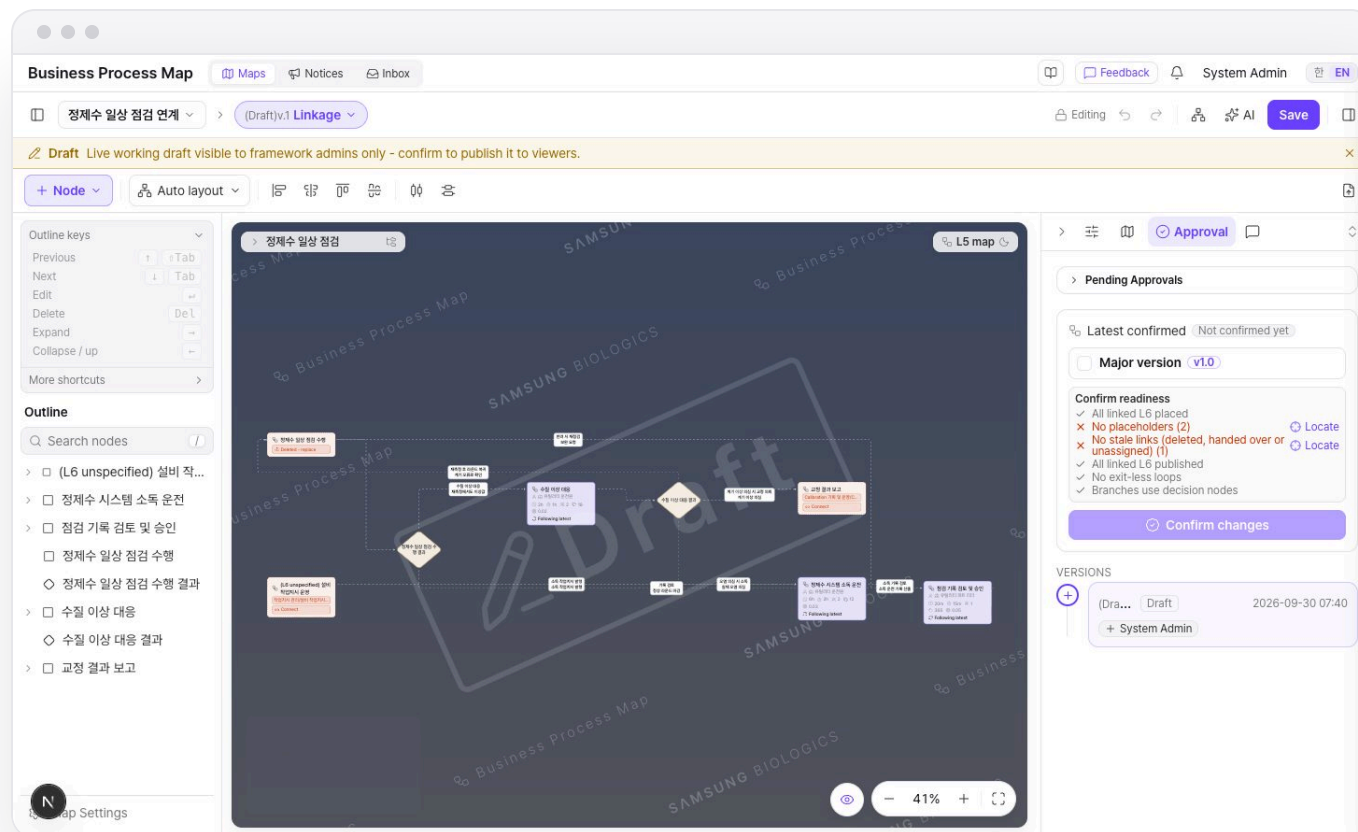
The Approval tab's **Confirm readiness** checklist. One failing gate disables Confirm.

Who confirms

The checkout holder if **direct admin** or sysadmin. Higher admins send a confirm request.

Draft visibility

Admins and sysadmins only. Everyone else lands on the latest confirmed snapshot.



Slot changes

Slot

Which L5 a map belongs to. Assign · unassign · move · hand off · delete, from the **Framework pill**.

Decider

That L5's direct admin (sysadmins too). Higher-category admins only request.

Move

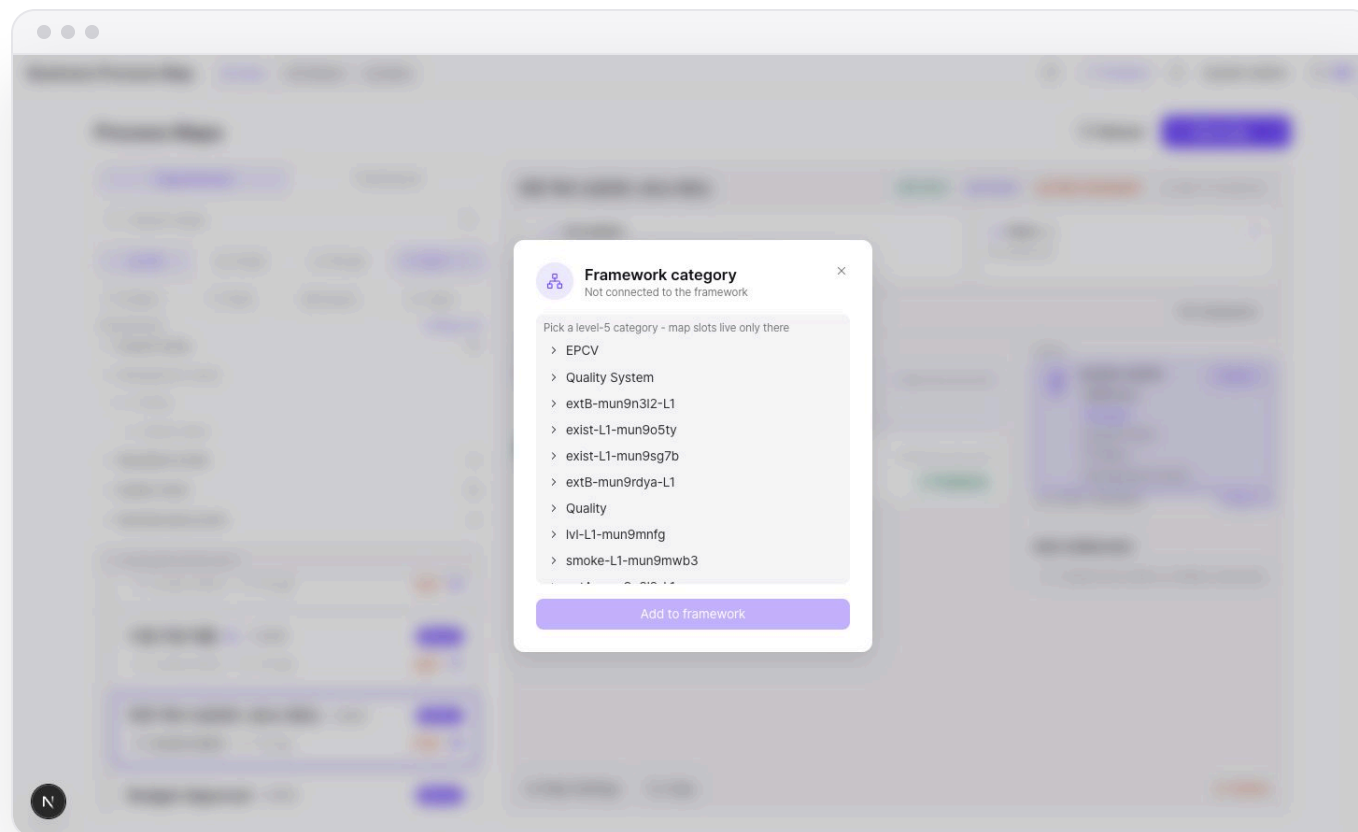
Both L5s' direct admins approve. An L5 without admins falls to sysadmin.

Request path

A direct-admin requester uses **Apply now**. Otherwise an `fw_slot` request lands in the Inbox and queue.

Fallback decider

Sysadmins decide from the queue. Map approvers cannot decide slot requests.



Status board

One table

Every L5 in scope. Path / Latest confirmed / Status (Ready · Blocked · No canvas) + **Open** .

Negative pills

Failing gates: Missing L6 · Placeholders · Stale links ·
Unpublished L6 · Exit-less loop · Direct fan-out.

Home link

The category summary card shows the same verdicts under Subtree confirmation.

Delegated scope

Delegated admins see their own subtree only.

Business Process Map

Maps

Notices

Inbox

Feedback

System Admin

한 EN

Settings

CONTENT

Notices

Manual

Knowledge base

AI chat

AI prompts

DIRECTORY

Employees

Departments

Catalogs

Orphaned refs

FRAMEWORK

Categories & Import

DATABASE

Tables

Batch jobs

APPROVALS

Approval Queue

GROUPS

User Groups

ANALYTICS

Dashboard

NH

Scheduled deletion

Categories & import

Re-imports overwrite category names/structure by code - the last delivery is the source of truth.

Manage

Status

Path	Latest confirmed	Status	
EPCV/Facility/계측 보전/Calibration 기획 및 운영/Calibration 수행 및 결과 보고	Not confirmed yet	No canvas	
EPCV/Facility/설비 보전/작업지시 관리/설비 작업지시 운영	Not confirmed yet	No canvas	
EPCV/Facility/유틸리티 운전/정제수 시스템 운전/정제수 일상 점검	Not confirmed yet	Blocked Placeholders (2) Stale links (deleted / handed over / unassigned) (1)	Open
Quality System/배치기록 검토/완제품 배치기록 검토/BRR 수행 및 종결/완제품 배치기록 검토 수행	Not confirmed yet	Ready	Open
Quality System/변경관리/제조 변경관리/변경 실행 및 완료/공정 변경관리 실행	Not confirmed yet	Ready	Open
Quality/품질관리(QC)/시험 운영/원제품 시험/원제품 출하 시험 관리	Not confirmed yet	Ready	Open
Quality/품질관리(QC)/시험 운영/원자재 시험/원자재 입고 시험 관리	Not confirmed yet	Blocked Placeholders (1)	Open
Quality/품질보증(QA)/품질 이벤트 관리/CAPA 관리/CAPA 실행 및 효과성 확인	Not confirmed yet	No canvas	
Quality/품질보증(QA)/품질 이벤트 관리/일탈 관리/시험 일탈-OOS 관리	Not confirmed yet	Blocked Placeholders (2)	Open
exist-L1-mun9o5ty/exist-L2-mun9o5ty/exist-L3-mun9o5ty/exist-L4-mun9o5ty/exist-L5-mun9o5ty	Not confirmed yet	Ready	Open

Interview import · format, dry run, governance

Flow

Interview JSON section → file-pill strip → Dry run → report → Apply. Sysadmin-only.

Format

0.4 (relations) · 0.5 (externalTasks). 0.3 rejected.
Re-runs are idempotent.

Two-column report

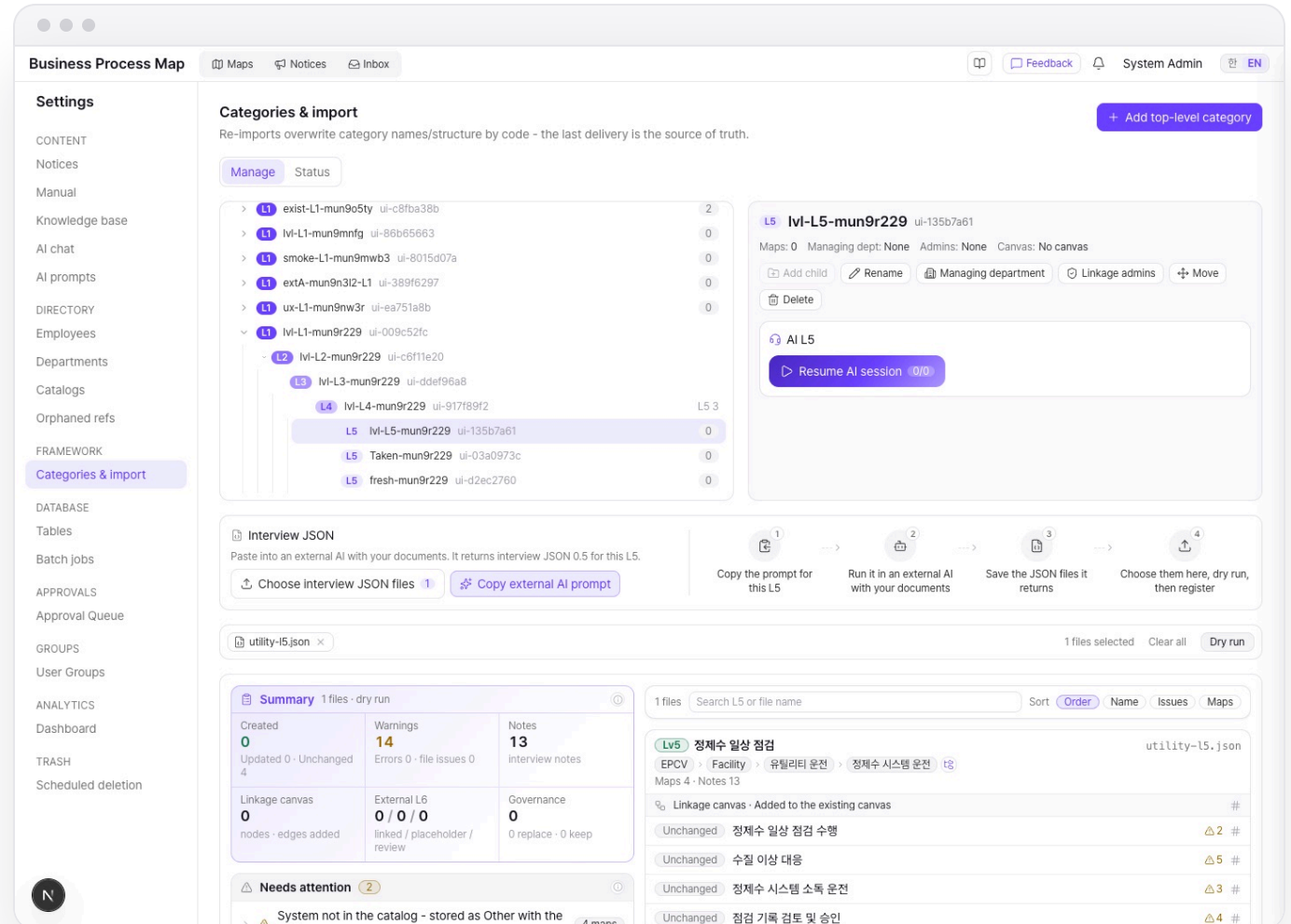
Left: Summary · Needs review · External L6 · admins · governance. Right: files → L5 canvas → maps.

Governance changes

On re-import, differing owner / department / approvers / notes become rows: Keep ↔ / Replace ↔.

Owner assignment

An empty or unknown owner makes the importing sysadmin interim owner (Owner unconfirmed).



Interview import · landing rules, placeholders, notes

Department path

Exact → leading levels dropped → unique suffix → mirror chain. No match registers the path as-is.

Landing

Activities become nodes, edges connectors. Exclusive → decision, parallel → parallel exit, self-loop → repeat branch.

Fan-out

Two or more outgoing edges that are not all parallel get a branch node after the activity.

Fields and notes

Conditions, times and systems land as fields, originals kept as source notes. Systems matched to the catalog.

Automatic

Exact output = input matches link as IO, horizontal auto-layout, a draft right after publish.

L5 canvas

Built or extended from top-level L6 flows. Redelivery only adds nodes.

Notes

Exceptions, VOC and regulatory basis become map notes; L5 entry / flow become category notes.

③ Connections · ④ Register

AI proposal

Entering Connections proposes the L6 flow at once. Plan predecessors are the backbone.

Full-width canvas

L5 linkage canvas with a floating feedback chat. Drag, right-click branches, auto-save.

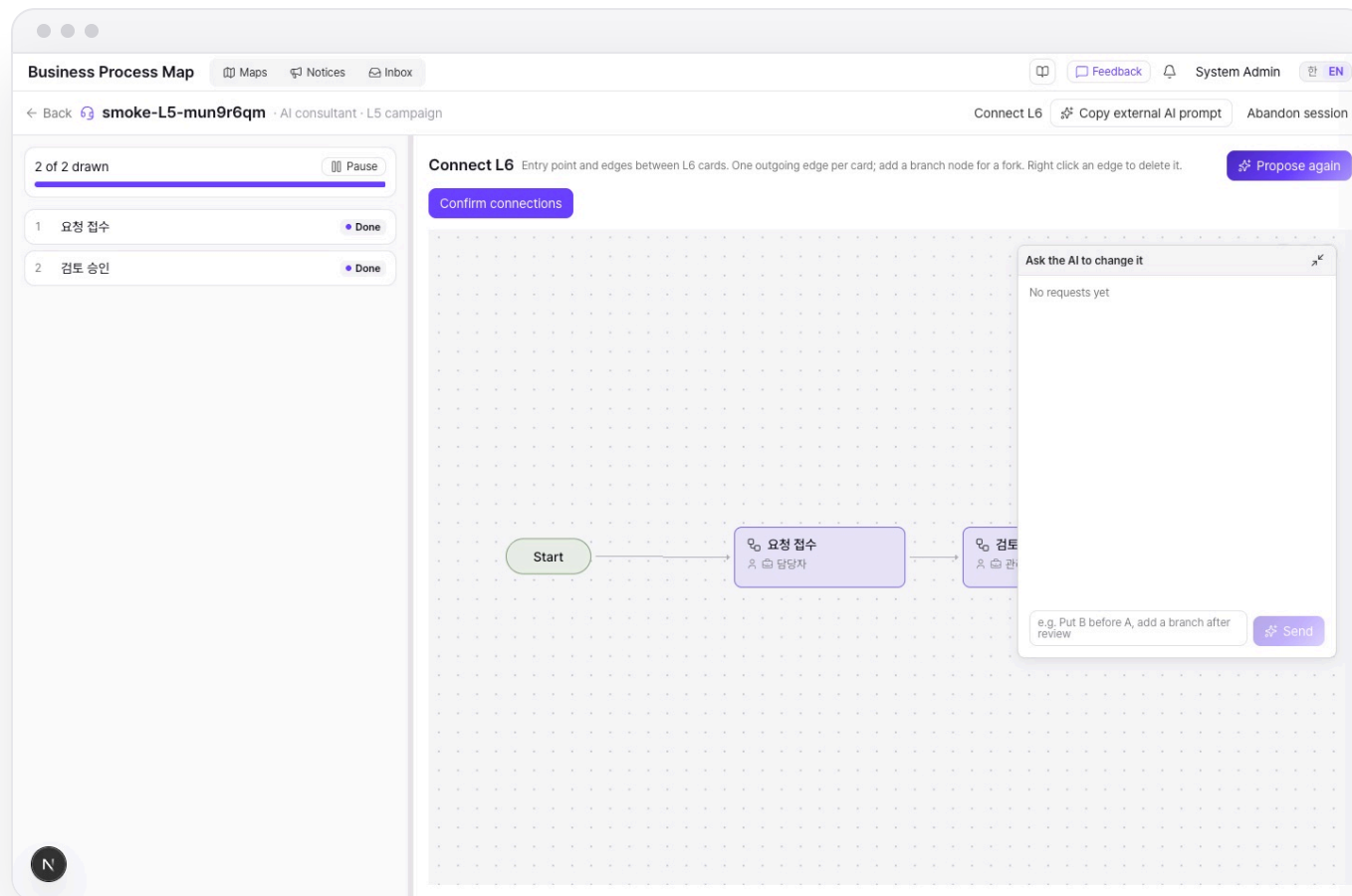
One outgoing connector

A new one marks the old in red dashes and replaces it.

Propose again restarts.

Register

Confirm connections → the import screen with an automatic Dry run. Maps publish, the canvas stays draft.



② Questionnaire · settling ambiguity

Per card

Decisions · exceptions · branches → activities → basics → I/O. A one-line reason per question.

Answer rules

All choice questions are required. Free text starts blank;
[AI suggestion] types it in.

Placeholder

No detail needed yet? **Create as placeholder**. Draw again with AI before registration.

Background drawing

Submitting draws that card's flow while the next questionnaire is prepared.

The screenshot displays the 'Business Process Map' interface. At the top, there are navigation tabs: 'Maps', 'Notices', and 'Inbox'. The main header shows the user 'ux-L5-mun9s6fk' and the role 'AI consultant · L5 campaign'. On the right, there are links for 'Feedback', 'System Admin', and a language selector '한 EN'.

The main content area is divided into several sections:

- Left Panel:** A list of cards with a '0 of 2 drawn' indicator and a 'Pause' button. The cards are:
 - 1. 요청 접수 (Ready)
 - 2. 검토 승인 편집 (Ready)
- Main Content Area:**
 - 1. 요청 접수:** A section for 'Decisions, exceptions and branches' containing:
 - 4. 시작 조건: 첨부에서 시작 시점이 없음, 요청서 도착
 - 5. 완결성 판정에서 미비하면?: 완결성 판정에서 미비될 때의 경로가 자료에 없음. Buttons: '보완 요청 후 재판정', '반려 종료'.
 - Activities:** A list of activities:
 - 1. 활동 순서
 - ✓ 요청 확인
 - ✓ 완결성 판정
 - ✓ 접수 등록
 - Basics:**
 - 2. 담당 역할: '담당자' and '관리자' buttons.
 - 3. 사용 시스템: 'ERP' (checked) and '메일' (unchecked).
 - Inputs and outputs:**
 - 6. 입력물: A text input field with an 'AI suggestion' button.
 - 7. 산출물: A text input field with an 'AI suggestion' button.

At the bottom, there is a note: 'Submitted cards cannot be reopened. Blank text answers stay unanswered.' and a 'Review answers' button.

① Plan · cards, stages, external L6

Propose L6 cards

Brief and attachments → cards. Three columns: purpose | stage rows (one row = parallel) | card details.

Stages

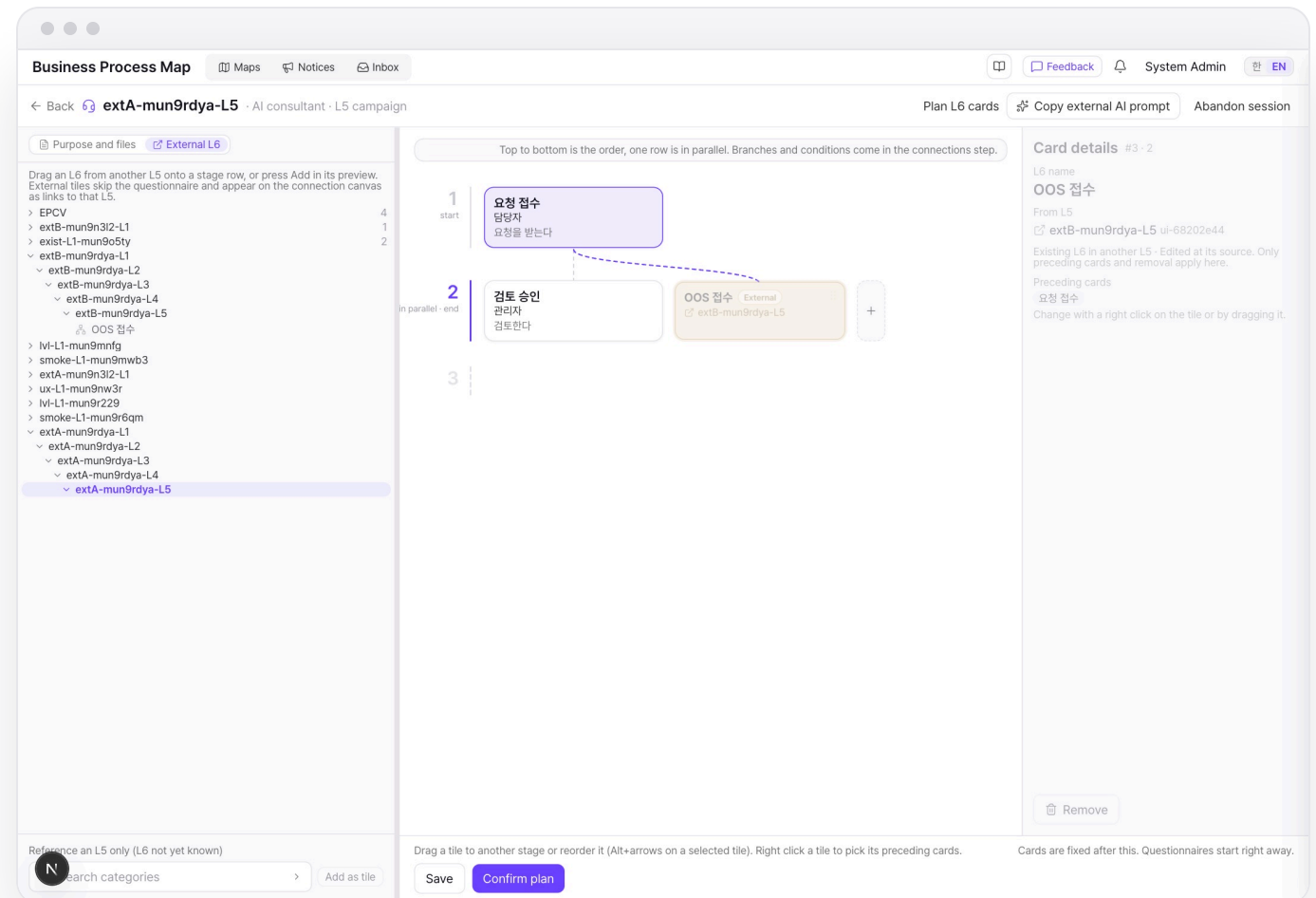
Drag tiles between rows, reorder, open a new stage.
Predecessors via tile **right-click**.

External L6

Tab or right-click an empty spot. Another L5's L6 as an external tile (connections only).

Pending removal

Removed cards stay as red tiles, restorable. **Lock** starts questionnaire prep.



Filling an L5 with AI · entry & sessions

Entry

Select a row in the category tree → level-specific **tile actions** in the detail panel. Sysadmin-only.

Per level

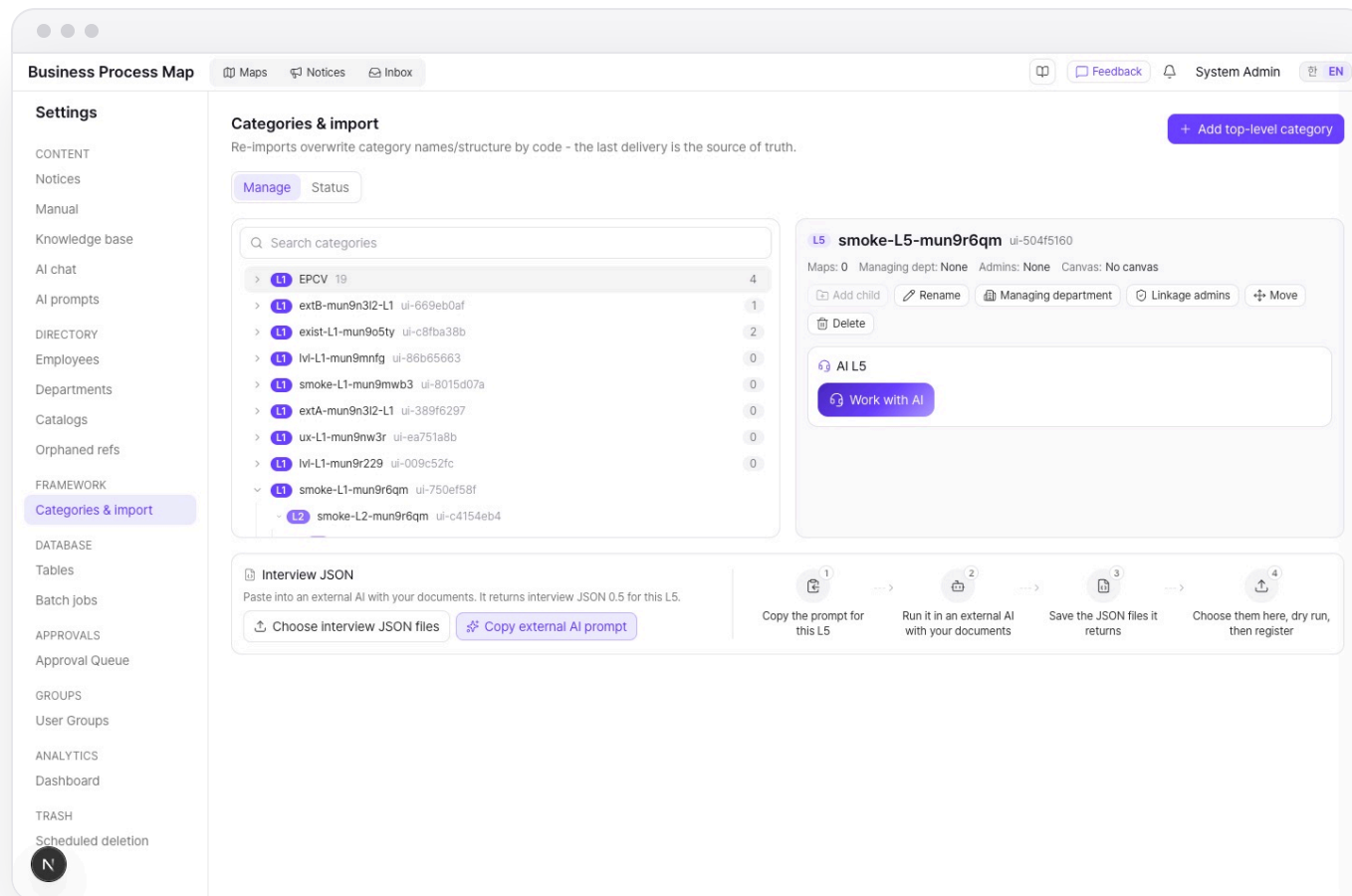
L1–L3 Go down · L4 **Create L5 and start with AI** · L5 **Work with AI** / Resume.

Sessions

Stored in the DB. Leaving keeps them, resume from that L5 row. **Copy external AI prompt** round-trips.

Four steps

① Plan → ② Questionnaire → ③ Connections → ④ Register. Registered L6 load first as **Existing**.



06

Database & Backups

The table viewer, notification purge, batch status, and daily backups.

The table viewer

Read-only

Server-side paging, sorting, filtering. It never writes.

CSV export

All rows with the current sort / filter.

Login records

`login_records`. One row per user per day, the audit view.

Notification purge

Selecting `notifications` adds a range purge. Grouped preview, then hard delete.

Retention

Notifications keep the most recent 100 per person.

The screenshot shows the 'Business Process Map' interface. On the left is a sidebar with a 'Settings' section and a 'DATABASE' section. The 'DATABASE' section has a 'Tables' sub-section highlighted. The main area displays a grid of database tables with their names and row counts. The 'notifications' table is selected and highlighted in purple. Below the grid, the 'notifications' table is expanded, showing a table with columns: id, recipient, type, map_id, version_id, message, payload, read, and created_at. The table contains 3 rows of data.

id	recipient	type	map_id	version_id	message	payload	read	created_at
1	admin.kim	review_requested	14	75	user.lee 님이 'As-Is' 승인을 요청했습니다.		false	2026-08-30T23:59:10.421538
2	admin.kim	review_requested	15	76	user.park 님이 'As-Is' 승인을 요청했습니다.		false	2026-08-30T23:59:10.422816
3	admin.kim	notice	17		새 공자: v2.4 승인 워크플로우 개편 안내가 등록되었습니다.		true	2026-08-30T23:59:10.424745

Batch job status

Per job

DB backup and HR sync. Last success and last failure with time and summary.

Signal

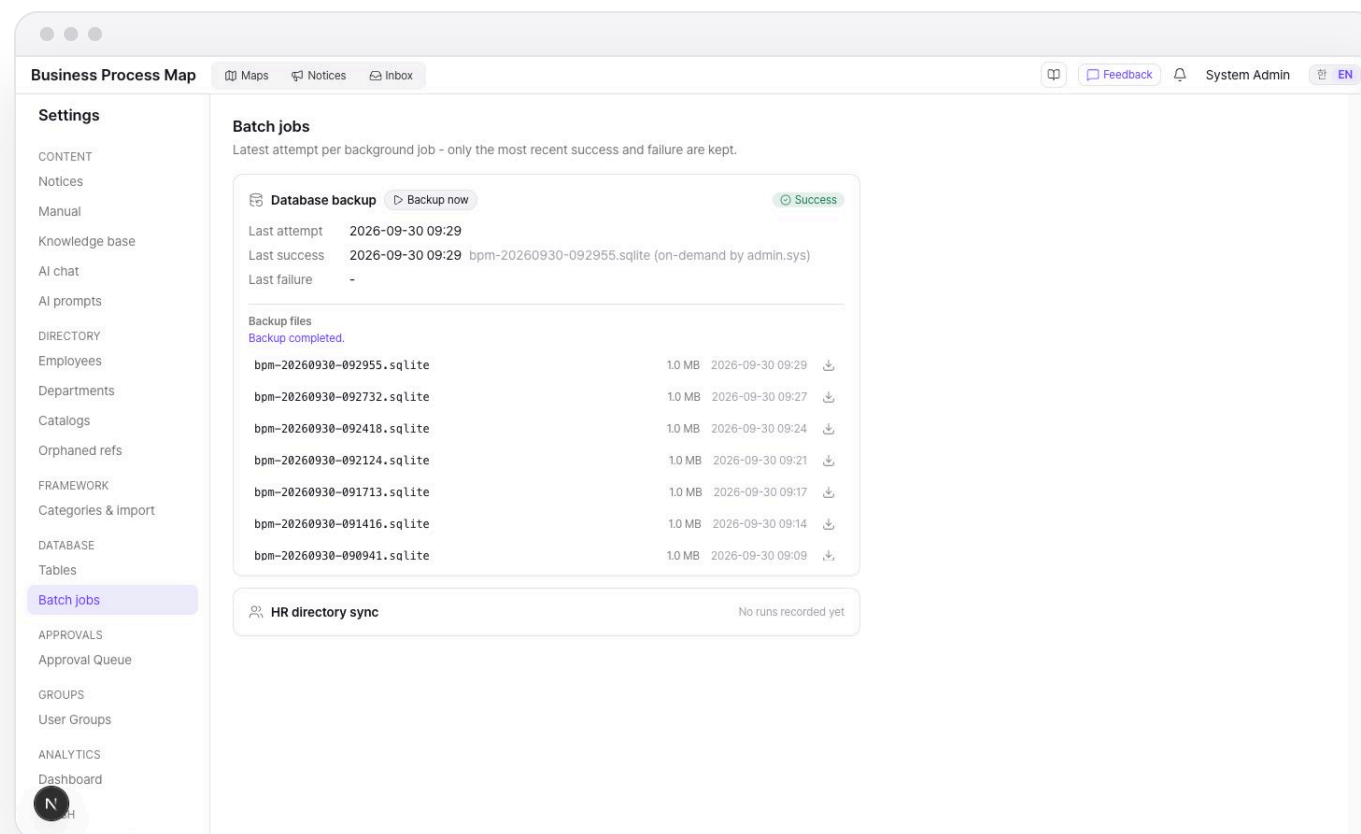
A failure newer than the last success means act.

Back up now

Outside the daily schedule. Production runs `pg_dump` via the sidecar, local copies sqlite.

Backup files

List of `${BACKUP_DIR}` with downloads for an off-server copy (sysadmin).



Daily automatic backups (prod)

db-backup sidecar

Daily after **04:00 KST**, plus once on boot if today's dump is missing.

Retention

`${BACKUP_DIR}` (default `./backups`) · `${BACKUP_RETENTION_DAYS}` (default 14).

Verification

Kept only after `pg_restore --list` passes. Outcomes land on the Batch jobs tab.

Off-server copies

Server disk only for now. Recovery runbook: `docs/deploy/backup.md`.

07

Approvals • Groups • Trash

The global queue, group approvals, restore, and instant purge.

The global approval queue

Everything pending

Group creation · permission downgrades · visibility · checkout transfers · **slot changes** (map · action · n/m).

Deciding

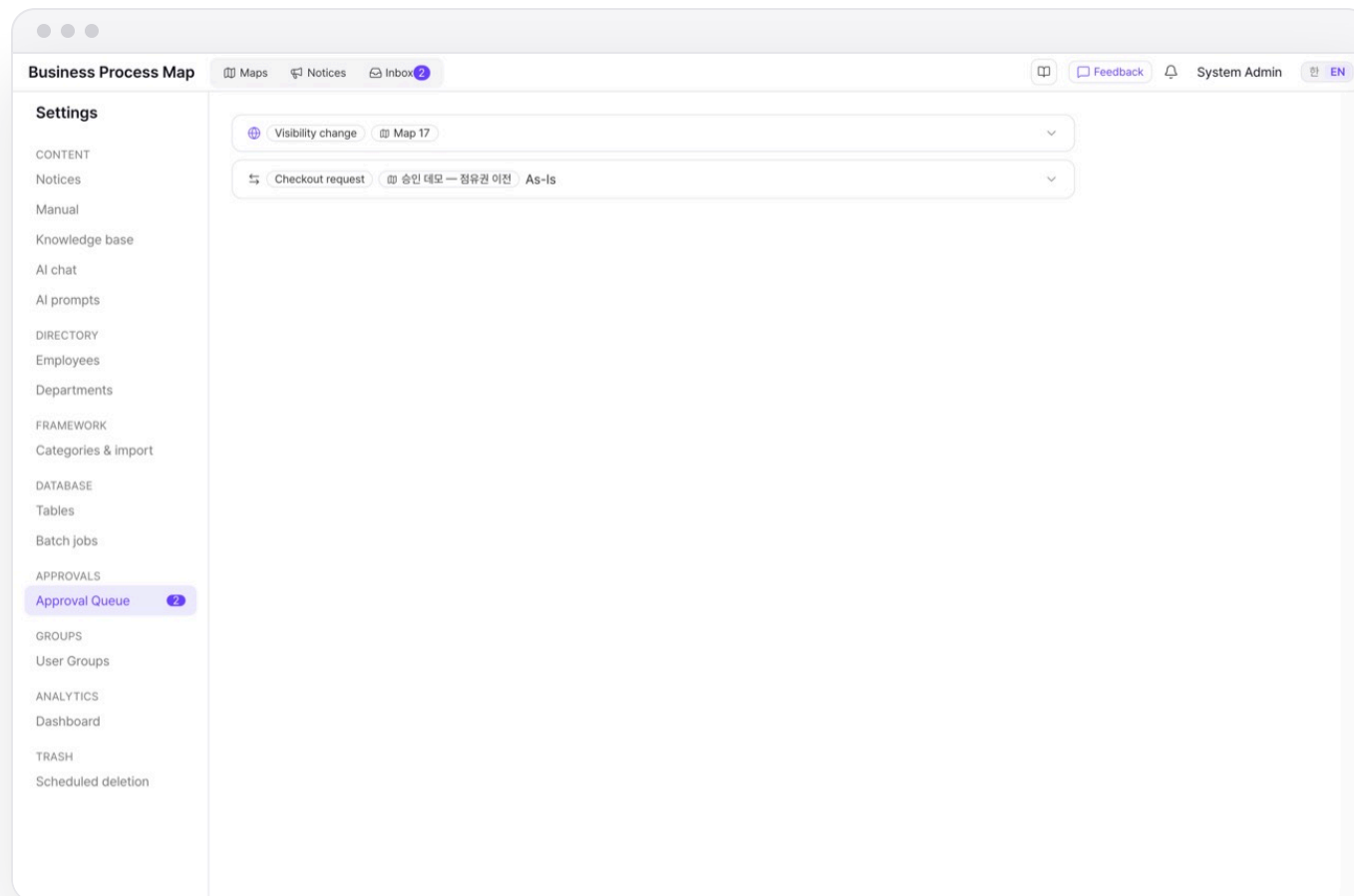
Approve / reject, no rejection reason. Map-scoped items are also decidable by the map's approvers.

Slot changes

Decided by the L5's direct admin. Sysadmins decide here as the fallback.

From the Inbox

Renames, SP registrations and confirm requests. A canvas failing its gates can't be approved.



User group administration

Request-based

Active after sysadmin approval.

Check

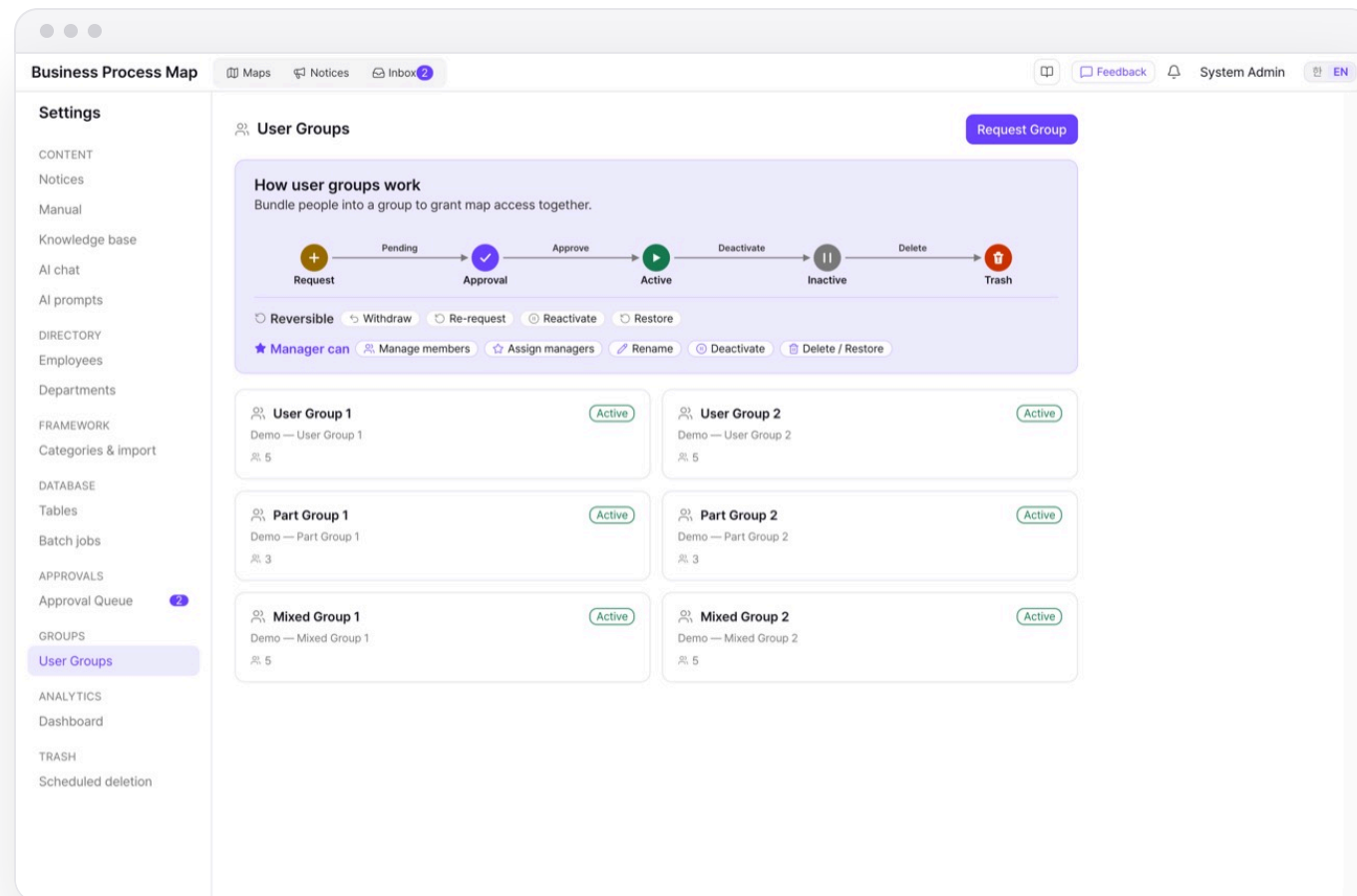
Name · members (2+) · managers (1+).

See all

Inactive and deleted groups included.

Deletion

7 days in the trash, then purged.



Trash & instant purge

Soft delete

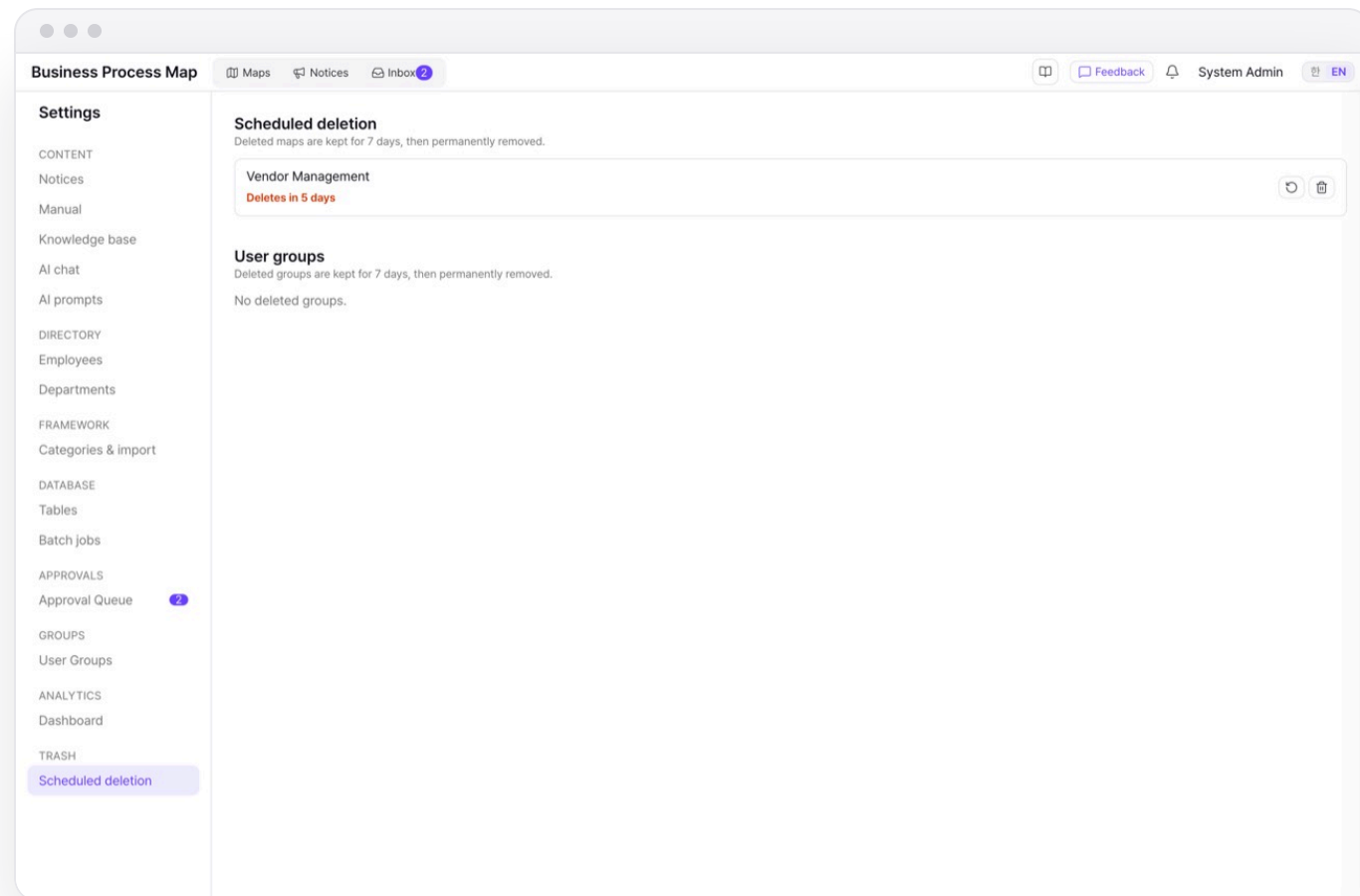
Maps and groups purge after 7 days. **Restore** until then.

Scope

Owners see their own, sysadmin sees everyone's trash.

Instant purge

Remove without the 7-day wait. Trash-state maps only,
no undo.



Version workflow · admin powers

Force checkout

Take the lock when the holder is away (unsaved context is lost). Checkouts never expire on their own.

Effective owner

Submit, publish and decide checkouts on any map.

Withdraw is the exception

Pending / Approved only by the submitter.

Approver reassignment

Recover approver-less maps via the forced-reassign flow.

Expiry

Publishing expires the previous version (terminal). Continue via Republish.

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Dashboard

Live operational metrics and delegated access.

The operations dashboard

Operations

Maps · published · in progress · trash, open comments, unread notifications, transfer requests.

Distribution

Version status and adoption by department (denominator in the Coverage sidebar).

Trends

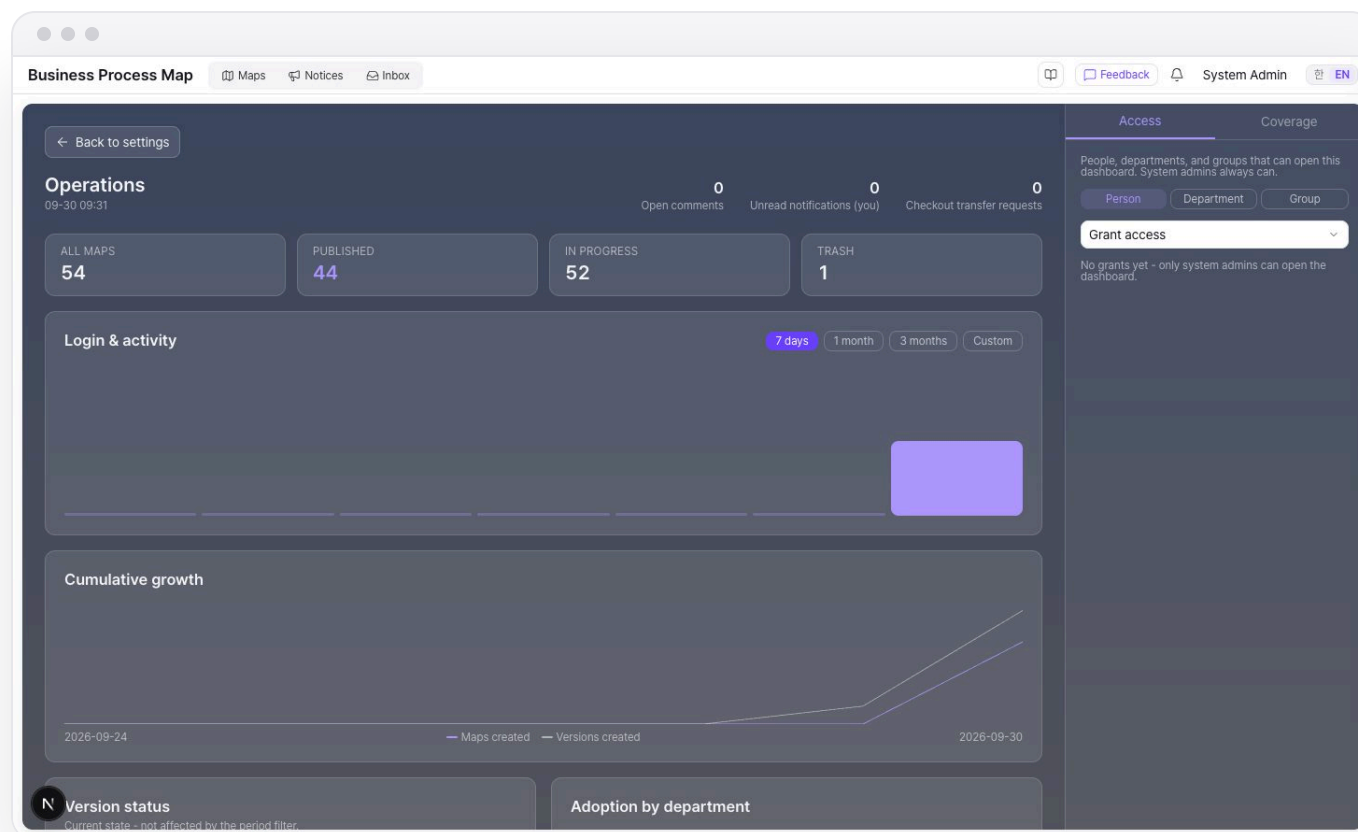
Login and activity, cumulative growth. 7d / 1m / 3m / custom period filter.

Events · AI

Recent version events, AI usage (7 / 30-day calls, top maps).

Access

Delegate viewing to people, departments, groups. Sysadmins always can.



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Configuration Reference

backend .env and compose variables. Changes need a backend restart.

Env · auth, permissions, AI

VARIABLE	EFFECT
BPM_SYSADMINS	Comma-separated sysadmin login IDs
AUTH_MODE	keycloak ldap dev . The frontend reads it at boot via GET /api/auth/mode
AUTH_JWT_SECRET / AUTH_JWT_TTL_HOURS	ldap session signing key (rotate = kill switch) · lifetime (default 8 h)
DEV_ENFORCE_PERMISSIONS	Enforce RBAC locally even with auth off
AI_BASE_URL / AI_MODEL	AI server (OpenAI-compatible) and model id. The in-house GPU uses just glm-5.2
AI_MAX_TOKENS / AI_TIMEOUT_SECONDS	Response token cap (too low = empty replies) · per-call timeout (120–180 s recommended)

Env · HR, backups, misc

VARIABLE	EFFECT
<code>N8N_HR_URL</code> / <code>N8N_HR_TOKEN</code>	HR webhook URL & secret. Sync activates only with both
<code>N8N_POSITION_URL</code>	EDW head-position webhook. Empty disables collection
<code>HR_SYNC_INTERVAL_HOURS</code>	Auto-sync interval (default 24, 0 = off). Keep 0 for the first migration
<code>HR_SYNC_DELETE_CAP_PCT</code>	Sync deletion cap % (default 20, 0 = off)
<code>BACKUP_DIR</code> / <code>BACKUP_RETENTION_DAYS</code>	Backup path (default ./backups, point at a NAS for off-server) · retention days (default 14)
<code>MANUAL_URL</code>	Editor-toolbar manual button. Hidden when empty

✱ Env changes need a backend restart (`--reload` does not re-read .env). All deployment values live in .env, never hardcoded.

Run it steady

Check the **Batch jobs tab** each morning, **AI prompt overrides** after deploys, and **Orphaned refs** after reorgs.